

Scalability & Future Plan

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	No authentication	Admin-style page is currently not protected by user roles.	High
2	External AI dependency	Live generation relies on Gemini API and network availability.	High
3	SQLite scaling limit	Local SQLite is not the intended store for high concurrency.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single/local prototype	Deploy behind an application server, add horizontal scaling and session-aware services when user demand grows.
2	Data Storage	SQLite local DB	Move to PostgreSQL/MySQL or another managed relational database.
3	Performance	AI generation is external and variable	Add caching, async job processing, rate limits and monitoring.
4	Security	API key in .env; no auth in current version	Add authentication, role-based authorization, secret manager and audit logs.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Authentication + User Profiles	1-2 months	Secure accounts and saved personal history
Phase 3	Progress Tracking + Workout History	2-3 months	Track adherence and prior plans
Phase 4	Exercise Media + Mobile / Cloud	3-6 months	Add demonstrations, device access and broader deployment